

CE 4011 Term Project — Proposal

Ali Utku Tekin — 2744076 MSc. Structural Engineering · Spring 2025–2026

Structural system and analysis method

The program analyses 2-D plane frames and trusses by the Direct Stiffness Method. Frame members are Euler–Bernoulli beam-columns with three DOFs per node (u_x , u_y , θ_z); truss members are pin-pin axial bars. The solver supports nodal loads, support settlements (partitioned solve), uniform distributed and in-span point loads on frames, axial-thermal and through-depth thermal-gradient loading, and internal hinges via Schur static condensation.

Planned advanced feature — free-vibration modal analysis

The program will assemble a global mass matrix from element-level consistent-mass matrices and material density ρ , then solve the generalised eigenvalue problem $(K - \omega^2 M)\phi = 0$ on the active free-DOF block via `scipy.linalg.eigh`. Outputs are natural angular frequencies ω_n , frequencies $f_n = \omega_n / 2\pi$, periods T_n , and mass-orthonormal mode shapes ϕ_n . Density is read in SI kg/m³ and converted internally to the solver's kN-m-s system.

Input / output

The existing `.txt` input format is preserved; density on `MATERIALS` becomes the only new field and defaults to 0 for backward compatibility. Outputs: free-DOF displacements, support reactions, element end-forces, an equilibrium-residual check, and — for modal runs — a frequency / period table plus the selected mode shape.

Visualisation (PyQt6 + matplotlib)

The GUI provides model creation, undeformed and (cubic-Hermite) deformed shapes, bending-moment / shear-force / axial-force diagrams, reaction arrows, and a modal-results pane with a mode selector and a deformation scale slider. A time-domain mode animation is included as optional polish.

Architecture and verification

Class layout: `StructuralModel`, `Node`, `Material`, `Section`, `Element2D` $\hookrightarrow$ `{FrameElement2D, TrussElement2D}`, with `assembler`, `solver`, `postprocessor`, `mass`, and `modal` modules and a separate `ModalResult` dataclass. Verification: `pytest` unit tests, closed-form checks (cantilever $PL^3/3EI$, simply-supported $PL/4$), modal references (clamped-free $\beta_n L = 1.875, 4.694, 7.855$; Chopra two-DOF shear building), and a mode-orthogonality check $\phi_i^T M \phi_j = 0$.